

GATE DA (Data Science & AI) Practice Question Bank - Set 7

3 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers three topics: Hypothesis Testing & Confidence Intervals, Ensembles (Bagging & Random Forest), and Probability (aptitude-level puzzles).

SAME HONESTY POLICY AS BEFORE: I searched for genuine GATE DA previous-year questions on all three topics. Hypothesis testing/confidence intervals and ensembles (bagging/random forest) are confirmed, syllabus-listed GATE DA topics (hypothesis testing was formally added to the syllabus for GATE DA 2026/2027; ensemble methods are listed under Machine Learning), but I could not locate a verbatim, publicly documented past GATE DA question on either topic with enough specific detail (exact numbers/wording) to reproduce reliably. So every question in those two topics is labelled [GATE-style] -- written to match real GATE DA difficulty and phrasing, but not a verbatim past question.

The third topic, Probability (aptitude-level puzzles), is deliberately written in the classic puzzle/riddle style used by general-aptitude question banks (dice, cards, coins, urns, truth-tellers) -- the same flavour you'll find on sites like IndiaBIX, Examveda, and Testbook's aptitude sections, and in the General Aptitude portion of any GATE paper. These are labelled [Aptitude-style, classic] since they are standard, well-established puzzle formats rather than quotes of a specific source.

Section B ("New Practice Questions") in every topic is freshly written practice, as always -- none of these are claimed to be past questions.

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Topic 1: Hypothesis Testing & Confidence Intervals

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style, confirmed core GATE DA syllabus topic]

Q1. [MCQ · 1 mark]

In hypothesis testing, the null hypothesis H_0 typically represents:

- (A) The claim we are trying to prove true
- (B) A default/status-quo statement of "no effect" or "no difference," which is tested for possible rejection
- (C) The alternative hypothesis being investigated
- (D) A statement that is always false

Answer: (B)

[GATE-style]

Q2. [MCQ · 1 mark]

A Type I error occurs when:

- (A) We fail to reject a false null hypothesis
- (B) We reject a true null hypothesis
- (C) We correctly reject a false null hypothesis
- (D) We accept a false alternative hypothesis

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

A Type II error occurs when:

- (A) We reject a true null hypothesis
- (B) We fail to reject a false null hypothesis
- (C) We correctly reject a false null hypothesis
- (D) We correctly fail to reject a true null hypothesis

Answer: (B)

[GATE-style]

Q4. [MCQ · 1 mark]

The significance level α of a hypothesis test represents:

- (A) The probability of a Type II error
- (B) The probability of rejecting H_0 when it is actually true (the Type I error rate)
- (C) The confidence level
- (D) The sample size

Answer: (B)

[GATE-style]

Q5. [NAT · 1 mark]

For a 95% confidence interval based on the normal distribution, the standard critical value $z_{\alpha/2}$ used is approximately (correct to 2 decimals):

Answer: 1.96

[GATE-style]

Q6. [NAT · 1 mark]

For a 99% confidence interval, the standard critical value $z_{\alpha/2}$ is approximately (correct to 3 decimals):

Answer: 2.576

[GATE-style]

Q7. [MCQ · 1 mark]

A z-test (rather than a t-test) is appropriate for testing a population mean when:

- (A) The population standard deviation is unknown and the sample size is small
- (B) The population standard deviation is known (or the sample size is large enough for it to be well-estimated)
- (C) The data is categorical
- (D) More than two groups are being compared

Answer: (B)

[GATE-style]

Q8. [NAT · 2 marks]

A sample of size $n=100$ has a known population standard deviation $\sigma=10$. Find the margin of error of the 95% confidence interval for the population mean (using $z=1.96$), correct to 2 decimals.

Answer: 1.96

Solution: Margin of error = $z \times \sigma / \sqrt{n} = 1.96 \times 10 / \sqrt{100} = 1.96 \times 10 / 10 = 1.96$.

[GATE-style]

Q9. [MCQ · 2 marks]

If a 95% confidence interval for a population mean is (48, 52), the CORRECT interpretation is:

- (A) There is a 95% probability that the true mean lies between 48 and 52
- (B) If the sampling process were repeated many times, about 95% of such constructed intervals would contain the true population mean
- (C) 95% of the individual data values lie between 48 and 52
- (D) The sample mean is definitely 50

Answer: (B)

Solution: This is the correct frequentist interpretation -- it describes the long-run behaviour of the interval-construction PROCEDURE across repeated sampling, not a probability statement about the (fixed) true parameter for this one particular interval.

[GATE-style]

Q10. [MCQ · 1 mark]

Which test is most appropriate for testing whether an observed categorical frequency distribution matches an expected distribution?

- (A) z-test
- (B) t-test
- (C) Chi-squared (χ^2) test
- (D) F-test

Answer: (C)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A sample of size $n=64$ has a known population standard deviation $\sigma=8$. Find the standard error of the mean ($\sigma/\sqrt{n}$).

Answer: 1

Solution: $8/\sqrt{64} = 8/8 = 1$.

[New Practice]

Q12. [NAT · 2 marks]

Using the previous question, find the 95% confidence interval's margin of error ($z=1.96$), correct to 2 decimals.

Answer: 1.96

Solution: Margin of error = $1.96 \times 1 = 1.96$.

[New Practice]

Q13. [MCQ · 1 mark]

Increasing the sample size n , all else being equal, will generally:

- (A) Increase the width of the confidence interval
- (B) Decrease the width of the confidence interval (a narrower, more precise estimate)
- (C) Have no effect on the confidence interval
- (D) Always double the width

Answer: (B)

[New Practice]

Q14. [MCQ · 1 mark]

Increasing the confidence level (e.g. from 95% to 99%), all else being equal, will:

- (A) Narrow the confidence interval
- (B) Widen the confidence interval (a higher confidence level requires a wider range)
- (C) Have no effect
- (D) Always halve the interval width

Answer: (B)

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE about the p-value in hypothesis testing?

- (A) A smaller p-value provides stronger evidence against the null hypothesis
- (B) If the p-value < significance level α , we reject the null hypothesis
- (C) The p-value is the probability that the null hypothesis is true
- (D) The p-value is computed assuming the null hypothesis is true

Answer: (A), (B), (D)

Solution: (C) is a common misconception -- the p-value is NOT the probability that H_0 is true; it is the probability of observing data at least as extreme as what was observed, ASSUMING H_0 is true.

[New Practice]

Q16. [NAT · 1 mark]

A t-test is performed with a sample of size $n=16$. Find the degrees of freedom ($n-1$) used for the t-distribution.

Answer: 15

[New Practice]

Q17. [MCQ · 1 mark]

As the degrees of freedom of a t-distribution increase (sample size grows), the t-distribution:

- (A) Becomes increasingly different from the standard normal distribution
- (B) Approaches the standard normal distribution
- (C) Becomes a chi-squared distribution
- (D) Becomes undefined

Answer: (B)

[New Practice]

Q18. [MCQ · 1 mark]

A one-tailed hypothesis test is used when:

- (A) The alternative hypothesis specifies a direction (e.g. "greater than" or "less than"), not just "not equal to"
- (B) There are two populations being compared, regardless of direction
- (C) The sample size is even
- (D) The population variance is unknown

Answer: (A)

[New Practice]

Q19. [NAT · 2 marks]

A confidence interval for a population proportion is constructed as $\hat{p} \pm z\sqrt{(\hat{p}(1-\hat{p}))/n}$. For $\hat{p}=0.4$, $n=225$, and $z=1.96$, find the margin of error (correct to 4 decimals).

Answer: 0.064

Solution: $\sqrt{(0.4 \times 0.6/225)} = \sqrt{(0.24/225)} = \sqrt{0.0010667} \approx 0.03266$. Margin of error = $1.96 \times 0.03266 \approx 0.0640$.

[New Practice]

Q20. [MCQ · 2 marks]

Which of the following correctly describes the relationship between hypothesis testing and confidence intervals for a population mean?

- (A) They are unrelated procedures
- (B) A $(1-\alpha)$ confidence interval contains exactly the values of the hypothesized mean that would NOT be rejected by a two-tailed test at significance level α
- (C) Confidence intervals can only be used for proportions, not means
- (D) Hypothesis tests always give wider ranges than confidence intervals

Answer: (B)

Topic 2: Ensembles – Bagging & Random Forest

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style, confirmed ML-syllabus topic]

Q1. [MCQ · 1 mark]

"Bagging" (Bootstrap Aggregating) primarily works by:

- (A) Training a single model on the full dataset, repeatedly
- (B) Training multiple models, each on a different bootstrap sample (a random sample with replacement) of the training data, and aggregating their predictions
- (C) Sequentially training models where each corrects the previous one's errors
- (D) Removing outliers before training a single model

Answer: (B)

[GATE-style]

Q2. [MCQ · 1 mark]

The primary motivation for bagging is to:

- (A) Reduce bias
- (B) Reduce variance, by averaging out the fluctuations of multiple independently-trained models
- (C) Increase training speed
- (D) Eliminate the need for a training set

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

A Random Forest extends bagging (applied to decision trees) by additionally:

- (A) Using boosting instead of bagging
- (B) Randomly selecting only a subset of features to consider at each split of each tree, further decorrelating the trees
- (C) Using only a single decision tree
- (D) Requiring labeled data to be duplicated exactly

Answer: (B)

[GATE-style]

Q4. [NAT · 2 marks]

In bootstrap sampling for bagging, a bootstrap sample of size n is drawn WITH replacement from a training set of size n . As $n \rightarrow \infty$, what fraction of the original data points is expected to be LEFT OUT of a single bootstrap sample? (Use the limit $(1 - 1/n)^n \rightarrow e^{-1}$; give the answer correct to 4 decimals.)

Answer: 0.3679

Solution: $e^{-1} \approx 0.3679$.

[GATE-style]

Q5. [MCQ · 1 mark]

The data points left out of a particular bootstrap sample (in bagging/Random Forest) are called:

- (A) Training data
- (B) Out-of-bag (OOB) samples

- (C) Validation folds
- (D) Support vectors

Answer: (B)

[GATE-style]

Q6. [MCQ · 1 mark]

Out-of-bag (OOB) samples in Random Forest/bagging are useful because they:

- (A) Must always be discarded
- (B) Can be used to estimate the model's generalization/test error WITHOUT needing a separate held-out validation set
- (C) Are used to train additional trees
- (D) Increase the bias of the ensemble

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

How are final predictions typically combined across the individual trees in a Random Forest for a CLASSIFICATION task?

- (A) Simple averaging of predicted probabilities / majority voting across trees
- (B) Only the first tree's prediction is used
- (C) Multiplying all tree predictions together
- (D) Taking the minimum prediction across trees

Answer: (A)

[GATE-style]

Q8. [MCQ · 1 mark]

How are final predictions typically combined across individual trees in a Random Forest for a REGRESSION task?

- (A) Majority voting
- (B) Averaging the numeric predictions of all the trees
- (C) Taking the maximum prediction
- (D) Concatenating all predictions

Answer: (B)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about ensemble methods (bagging/Random Forest) in general?

- (A) They typically reduce variance compared to a single base model (e.g. a single unpruned decision tree)
- (B) They can help mitigate overfitting relative to a single deep decision tree
- (C) Bagging trains its base models sequentially, each depending on the previous model's errors
- (D) Random Forest introduces additional randomness (feature subsampling) beyond plain bagging, to decorrelate trees

Answer: (A), (B), (D)

Solution: (C) is false -- that describes BOOSTING, not bagging. Bagging trains its base models independently and in parallel (each on its own bootstrap sample), not sequentially.

[GATE-style]

Q10. [MCQ · 2 marks]

Compared to a single deep, unpruned decision tree, a Random Forest generally has:

- (A) Higher variance and higher bias
- (B) Lower variance, with comparable or slightly higher bias
- (C) Exactly the same bias and variance
- (D) Higher variance and lower bias

Answer: (B)

Solution: Averaging over many decorrelated trees substantially reduces variance; bias typically changes little, or increases very slightly, compared to a single unpruned tree.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A Random Forest ensemble consists of 100 decision trees. For a binary classification task, 62 trees vote for class 1 and the rest vote for class 0. What is the ensemble's final predicted class (majority vote), as 0 or 1?

Answer: 1

Solution: 62 out of 100 > 50, so the majority vote is class 1.

[New Practice]

Q12. [MCQ · 1 mark]

If the individual base models (trees) in a bagging ensemble were perfectly CORRELATED (identical) with each other, the variance-reduction benefit of bagging would be:

- (A) Maximized
- (B) Completely lost -- averaging identical (fully correlated) models provides no variance reduction
- (C) Unaffected by correlation
- (D) Negative

Answer: (B)

[New Practice]

Q13. [MCQ · 1 mark]

Which of the following is a hyperparameter specific to Random Forest (beyond standard decision-tree hyperparameters like max depth)?

- (A) The learning rate
- (B) The number of features randomly considered at each split (often called "max_features" or "mtry")
- (C) The regularization coefficient λ
- (D) The kernel type

Answer: (B)

[New Practice]

Q14. [NAT · 2 marks]

A bootstrap sample of size $n=500$ is drawn with replacement from a training set of size 500. Using the approximation $e^{-1} \approx 0.3679$, approximately how many of the 500 original data points are expected to be LEFT OUT (out-of-bag)?

Answer: 184

Solution: $500 \times 0.3679 \approx 184$ (rounded to the nearest whole number).

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are advantages of Random Forests over a single decision tree?

- (A) Generally better generalization performance (lower test error), due to variance reduction
- (B) Built-in estimate of feature importance
- (C) Guaranteed to always have zero bias
- (D) Reduced risk of overfitting compared to one very deep, unpruned tree

Answer: (A), (B), (D)

Solution: (C) is false -- no model is guaranteed to have zero bias; Random Forests still carry some bias, though often comparable to a single deep tree.

[New Practice]

Q16. [MCQ · 1 mark]

Which of the following best distinguishes bagging from boosting?

- (A) Bagging trains models sequentially; boosting trains them independently in parallel
- (B) Bagging trains base models independently/in parallel on bootstrap samples; boosting trains them sequentially, with each new model focusing more on the previous models' errors
- (C) They are exactly the same technique
- (D) Bagging always uses decision trees; boosting never does

Answer: (B)

[New Practice]

Q17. [MCQ · 1 mark]

The number of trees in a Random Forest is generally increased (within reasonable computational limits) because:

- (A) It tends to reduce variance further and stabilize predictions, with diminishing returns, and does not typically cause overfitting by itself
- (B) More trees always cause severe overfitting
- (C) It decreases the training time
- (D) It has no effect on model performance at all

Answer: (A)

[New Practice]

Q18. [NAT · 1 mark]

A bagging ensemble uses 50 base classifiers (trees) for a binary classification problem. If 28 trees vote for class 0, how many trees vote for class 1?

Answer: 22

Solution: $50 - 28 = 22$.

[New Practice]

Q19. [MCQ · 2 marks]

Which of the following statements about feature importance in Random Forests is generally TRUE?

- (A) Feature importance cannot be computed for Random Forests
- (B) Feature importance can be estimated, e.g. by measuring the average decrease in impurity (or increase in OOB error when a feature's values are permuted) attributable to that feature across all trees
- (C) Feature importance is always identical for every feature
- (D) Feature importance requires a completely separate model

Answer: (B)

[New Practice]

Q20. [MCQ · 2 marks]

Which of the following is generally TRUE about applying bagging to a model that is already very stable/low-variance (e.g. simple linear regression), as opposed to a high-variance model like a deep decision tree?

- (A) Bagging provides the same large variance-reduction benefit regardless of the base model's stability
- (B) Bagging provides comparatively little benefit for already low-variance/stable models, since there is little variance left to reduce
- (C) Bagging always dramatically increases the model's bias
- (D) Bagging cannot be applied to linear regression at all

Answer: (B)

Topic 3: Probability - Aptitude-Level Puzzles

Section A — Old / Previous-Year-Style Questions (10)

[Aptitude-style, classic]

Q1. [MCQ · 1 mark]

In a simultaneous throw of two dice, find the probability of getting a total more than 7.

- (A) $5/12$
- (B) $1/2$
- (C) $1/3$
- (D) $7/12$

Answer: (A)

Solution: Favourable sums: 8,9,10,11,12 with counts 5,4,3,2,1 respectively (out of 36 total outcomes), giving 15 favourable outcomes. $P = 15/36 = 5/12$.

[Aptitude-style, classic]

Q2. [NAT · 2 marks]

A bag contains 6 white and 4 black balls. Two balls are drawn at random. Find the probability that both balls are of the same colour (correct to 4 decimals).

Answer: 0.4667

Solution: Same colour = both white or both black. $C(6,2) + C(4,2) = 15 + 6 = 21$. Total ways = $C(10,2) = 45$. $P = 21/45 = 7/15 \approx 0.4667$.

[Aptitude-style, classic]

Q3. [MCQ · 1 mark]

Three coins are tossed simultaneously. What is the probability of getting at least two heads?

- (A) $1/2$
- (B) $3/8$
- (C) $1/4$
- (D) $5/8$

Answer: (A)

Solution: Outcomes with ≥ 2 heads: HHH, HHT, HTH, THH -- 4 out of 8 total. $P = 4/8 = 1/2$.

[Aptitude-style, classic]

Q4. [NAT · 1 mark]

A card is drawn from a well-shuffled deck of 52 cards. Find the probability that it is neither a king nor a queen (correct to 4 decimals).

Answer: 0.8462

Solution: Kings + queens = $4 + 4 = 8$. Remaining = $52 - 8 = 44$. $P = 44/52 = 11/13 \approx 0.8462$.

[Aptitude-style, classic riddle]

Q5. [MCQ · 1 mark]

What is the probability that two people chosen at random were NOT born on the same day of the week (assume all 7 days are equally likely)?

- (A) $6/7$
- (B) $1/7$

- (C) $\frac{1}{2}$
(D) $\frac{5}{7}$

Answer: (A)

Solution: $P(\text{same day}) = \frac{1}{7}$. $P(\text{not the same day}) = 1 - \frac{1}{7} = \frac{6}{7}$.

[Aptitude-style, classic]

Q6. [NAT · 2 marks]

In a class, the probability of a student passing Maths is 0.6, passing English is 0.5, and passing both is 0.3. Find the probability of passing at least one subject.

Answer: 0.8

Solution: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.3 = 0.8$.

[Aptitude-style, classic]

Q7. [MCQ · 2 marks]

A speaks the truth 3 out of 4 times, and B speaks the truth 4 out of 5 times. Find the probability that they CONTRADICT each other in stating the same fact.

- (A) $\frac{7}{20}$
(B) $\frac{3}{20}$
(C) $\frac{1}{2}$
(D) $\frac{13}{20}$

Answer: (A)

Solution: $P(\text{contradict}) = P(A \text{ true, } B \text{ false}) + P(A \text{ false, } B \text{ true}) = \left(\frac{3}{4}\right)\left(\frac{1}{5}\right) + \left(\frac{1}{4}\right)\left(\frac{4}{5}\right) = \frac{3}{20} + \frac{4}{20} = \frac{7}{20}$.

[Aptitude-style, classic]

Q8. [NAT · 2 marks]

From a group of 4 men and 5 women, a committee of 3 is formed at random. Find the probability that the committee has exactly 2 women and 1 man (correct to 4 decimals).

Answer: 0.4762

Solution: $\frac{C(5,2) \times C(4,1)}{C(9,3)} = \frac{(10 \times 4)}{84} = \frac{40}{84} = \frac{10}{21} \approx 0.4762$.

[Aptitude-style, classic]

Q9. [MCQ · 2 marks]

A bag contains 5 red and 3 green balls. Two balls are drawn one after another WITHOUT replacement. Find the probability that the first is red and the second is green.

- (A) $\frac{15}{56}$
(B) $\frac{5}{8}$
(C) $\frac{3}{7}$
(D) $\frac{15}{28}$

Answer: (A)

Solution: $P = \left(\frac{5}{8}\right) \times \left(\frac{3}{7}\right) = \frac{15}{56}$ (the denominator shrinks to 7 after the first, non-replaced draw).

[Aptitude-style, classic]

Q10. [MCQ · 1 mark]

A fair coin is tossed 3 times. What is the probability of getting AT LEAST one tail?

- (A) $\frac{7}{8}$

- (B) $1/8$
- (C) $3/8$
- (D) $1/2$

Answer: (A)

Solution: P(no tails at all, i.e. all heads) = $1/8$. P(at least one tail) = $1 - 1/8 = 7/8$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

Two dice are thrown together. Find the probability that the sum is a multiple of 4 (correct to 4 decimals).

Answer: 0.25

Solution: Multiples of 4 achievable: 4 (3 ways), 8 (5 ways), 12 (1 way) = 9 favourable outcomes out of 36. $P = 9/36 = 0.25$.

[New Practice]

Q12. [MCQ · 1 mark]

A card is drawn from a standard deck. Find the probability that it is a face card (King, Queen, or Jack).

- (A) $3/13$
- (B) $1/4$
- (C) $4/13$
- (D) $1/13$

Answer: (A)

Solution: Face cards = 12 out of 52 = $12/52 = 3/13$.

[New Practice]

Q13. [NAT · 1 mark]

A bag has 4 red, 5 blue, and 3 green balls (12 total). One ball is drawn at random. Find the probability that it is NOT green (correct to 4 decimals).

Answer: 0.75

Solution: Not green = 9 out of 12. $P = 9/12 = 0.75$.

[New Practice]

Q14. [MCQ · 1 mark]

Two unbiased coins are tossed. What is the probability of getting at most one head?

- (A) $3/4$
- (B) $1/2$
- (C) $1/4$
- (D) 1

Answer: (A)

Solution: Outcomes: HH, HT, TH, TT. At most 1 head: HT, TH, TT -- 3 out of 4. $P = 3/4$.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE about mutually exclusive events A and B (with $P(A) > 0$, $P(B) > 0$)?

- (A) $P(A \cap B) = 0$
- (B) $P(A \cup B) = P(A) + P(B)$
- (C) A and B must also be independent
- (D) Both A and B cannot occur simultaneously in a single trial

Answer: (A), (B), (D)

Solution: (C) is false -- mutually exclusive events with positive probabilities can never be independent, since $P(A \cap B) = 0 \neq P(A)P(B) > 0$.

[New Practice]

Q16. [NAT · 1 mark]

In a lottery, there are 10 prizes and 25 blanks. A lottery ticket is drawn at random. Find the probability of getting a prize (correct to 4 decimals).

Answer: 0.2857

Solution: $P = 10/35 = 2/7 \approx 0.2857$.

[New Practice]

Q17. [MCQ · 2 marks]

A problem is given to three students A, B, C whose probabilities of solving it are $1/2$, $1/3$, and $1/4$ respectively. What is the probability that the problem is solved (i.e. at least one of them solves it)?

- (A) $3/4$
- (B) $1/4$
- (C) $1/2$
- (D) $11/12$

Answer: (A)

Solution: $P(\text{none solve}) = (1 - 1/2)(1 - 1/3)(1 - 1/4) = (1/2)(2/3)(3/4) = 6/24 = 1/4$. $P(\text{at least one solves}) = 1 - 1/4 = 3/4$.

[New Practice]

Q18. [NAT · 2 marks]

A box contains 3 blue, 2 red, and 5 yellow marbles. Two marbles are drawn without replacement. Find the probability that both are yellow (correct to 4 decimals).

Answer: 0.2222

Solution: $C(5,2)/C(10,2) = 10/45 = 2/9 \approx 0.2222$.

[New Practice]

Q19. [MCQ · 2 marks]

A committee of 5 is to be formed from 6 men and 4 women, chosen at random. What is the probability that the committee has at least one woman?

- (A) $41/42$
- (B) $1/42$
- (C) $5/9$
- (D) $4/9$

Answer: (A)

Solution: $P(\text{all men}) = C(6,5)/C(10,5) = 6/252 = 1/42$. $P(\text{at least one woman}) = 1 - 1/42 = 41/42$.

[New Practice]

Q20. [NAT · 2 marks]

A fair die is rolled twice. Find the probability that the second roll is strictly greater than the first roll (correct to 4 decimals).

Answer: 0.4167

Solution: Counting favourable pairs (first,second) with $\text{second} > \text{first}$: for first=1, 5 options; first=2, 4; first=3, 3; first=4, 2; first=5, 1; first=6, 0. Total = $5+4+3+2+1+0 = 15$, out of 36. $P = 15/36 = 5/12 \approx 0.4167$.
